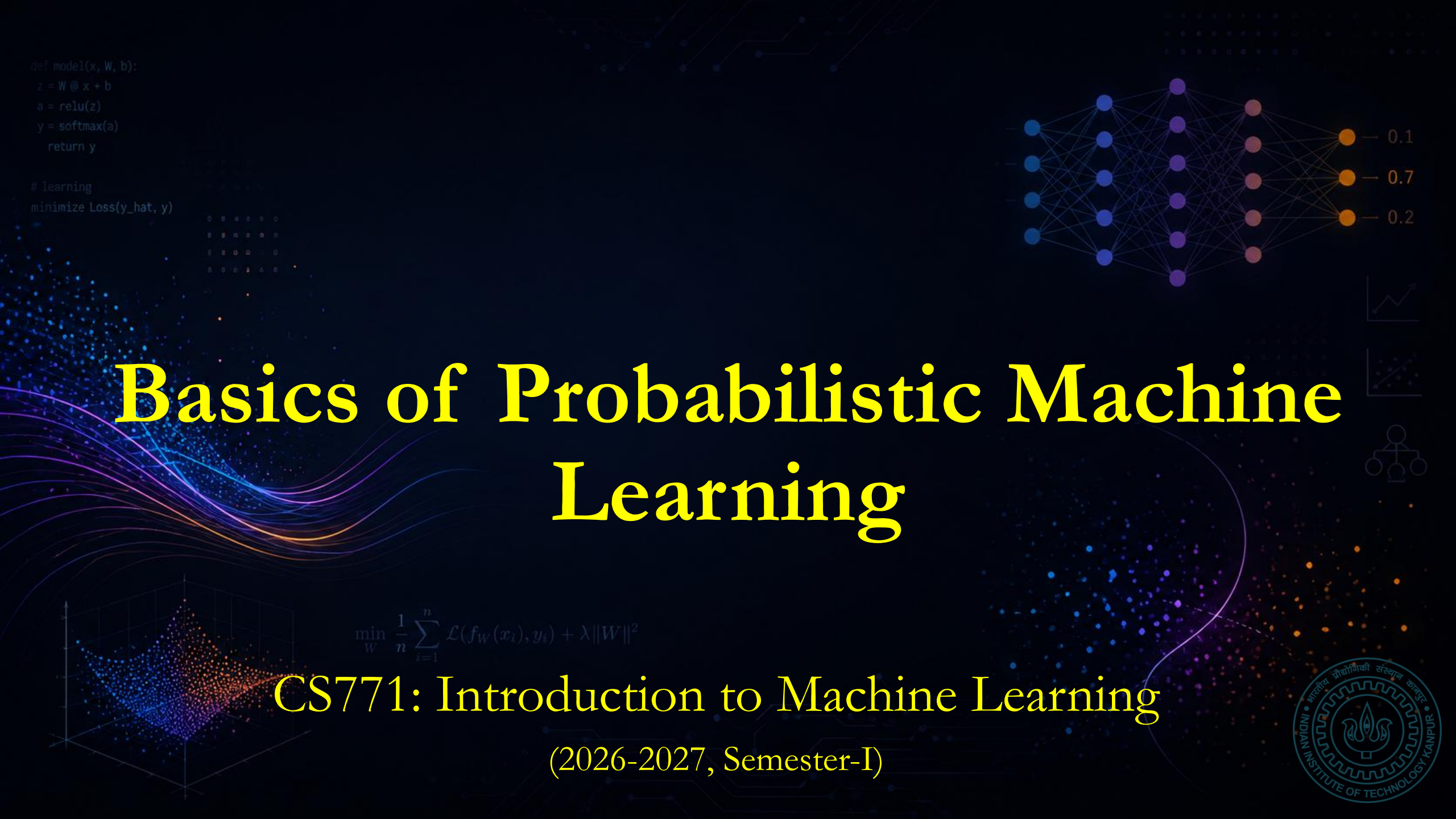


Basics of Probabilistic Machine Learning

CS771: Introduction to Machine Learning

(2026-2027, Semester-I)

Why Probabilistic Modeling for ML?

A probabilistic model helps us making probabilistic predictions

Deterministic
$\hat{y} = f(\boldsymbol{x}; \boldsymbol{\theta})$

vs

Probabilistic
$p(y \mid \boldsymbol{x}, \boldsymbol{\theta})$

Often desirable to not just report a single prediction but probabilities of each possible outcome, or the range in which the true output lies

A probabilistic model helps learn the probability distribution that models some observed data $\mathcal{D}$ (say a collection of images or text) and can be used to generate novel synthetic data (generative AI)

$$x \sim p(\boldsymbol{x}|\mathcal{D})$$

A probabilistic model can help account for uncertainty in the parameters that we have estimated (important if we have limited/noisy data)

Using appropriate probability distributions for the data/parameters

A probabilistic model can help incorporate assumptions about data/parameteters

A Simple Probabilistic Learning Problem

Consider a manufacturing plant producing items

$$y = \begin{cases} 1, & \text{item is functioning,} \\ 0, & \text{item is not functioning (defective)} \end{cases}$$



For now, we won't assume that any specific features $\boldsymbol{x}$ of the item are used to predict the item's functioning

If features $\boldsymbol{x}$ are also available, we can actually estimate $p(y = 1|x, \boldsymbol{w})$ using weights $\boldsymbol{w}$. Will see that later

Want to estimate $\theta \in [0,1]$, s.t.

$$p(y = 1|\theta) = \theta$$

$$p(y = 0|\theta) = 1 - \theta$$

Here θ is the the probability that a randomly inspected item from the plant will be functioning.

Data: Inspection outcomes $\mathcal{D} = \{y_1, \ldots, y_N\}$, of N items, $y_n \in \{0,1\}$

Same setup in many other problems: heads vs tails, click vs no-click, component failure vs no failure, or positive vs negative class labels.

A Bernoulli Model for Binary Observations

Binary observations can be naturally modeled using a Bernoulli

$$y_n|\theta \sim \text{Bernoulli}(\theta)$$

$\theta \in (0,1)$ is called the "bias" or the probability of "success" of the Bernoulli trial

Bernoulli's probability mass function (PMF)

An appropriate probability distribution for the binary observations in the manufacturing plant problem

$$p(y_n|\theta) = \theta^{y_n}(1-\theta)^{1-y_n}$$

$$p(y_n = 1|\theta) = \theta$$

$$p(y_n = 0|\theta) = 1-\theta$$

Assume that given θ (a.k.a. "conditioned" on θ), the observations $\mathcal{D} = \{y_1, \ldots, y_N\}$ are independent and identically distributed (i.i.d.)

$$y_1, \ldots, y_N|\theta \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$$



Population Probability versus Observed Fraction

The true probability of success $\theta = P(y = 1)$ is a property of data "population" or data-generating process.

Observed fraction of positives in a finite dataset $\mathcal{D} = \{y_1, \ldots, y_N\}$,

$$\hat{\theta} = \frac{N_1}{N}, \qquad N_1 = \sum_{n=1}^{N} y_n ,$$

is a property of the particular data sample $\mathcal{D}$ we happened to observe

Important: $\hat{\theta}$ is not the same as the true probability of success θ

$\hat{\theta}$ is simply a data-based estimate of the true θ



Likelihood vs Probability

Consider the Bernoulli model for each observation

$$p(y_n \mid \theta) = \theta^{y_n}(1-\theta)^{1-y_n}$$

Important nomenclature: The quantity $p(y_n \mid \theta)$ has two interpretations

(1) Probability: Fix θ and vary $y_n \in \{0,1\}$

(2) Likelihood: Fix y_n and vary $\theta \in (0,1)$

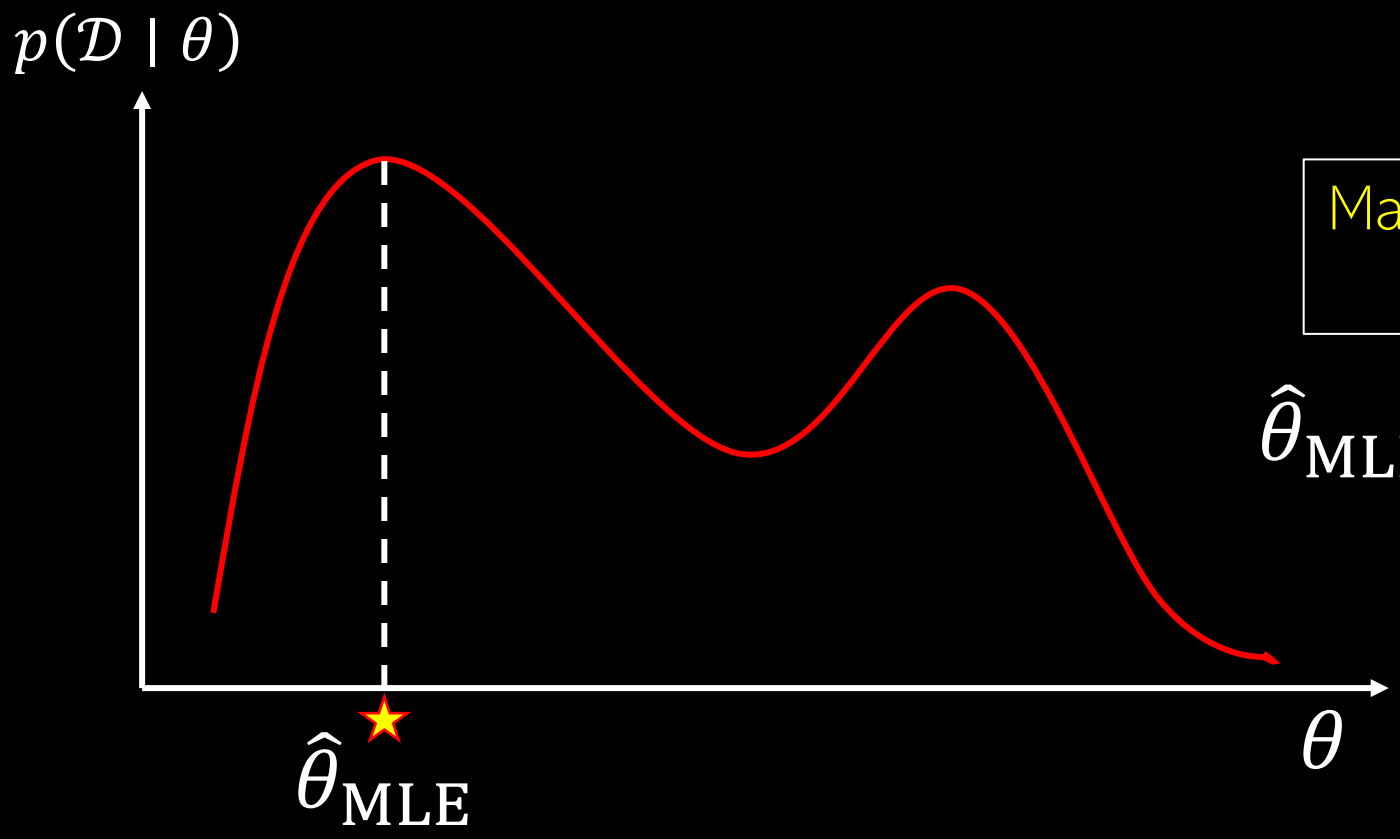
Thus, likelihood is a function $L(\theta; y_n)$ of the parameter θ when y_n has been fixed (observed)

Likelihood for the observed dataset (assuming i.i.d.) $\mathcal{D} = \{y_1, \ldots, y_N\}$

$$L(\theta; \mathcal{D}) = p(\mathcal{D} \mid \theta) = \prod_{n=1}^{N} p(y_n \mid \theta) = \theta^{\sum_{n=1}^{N} y_n}(1-\theta)^{N-\sum_{n=1}^{N} y_n}$$

Using Likelihood for Parameter Estimation

If using only observed data $\mathcal{D}$, the likelihood $L(\theta; \mathcal{D}) = p(\mathcal{D} \mid \theta)$ can help us estimate θ by comparing different θ values to find the "best" θ



MLE is basically an optimization problem (maximize the log likelihood or minimize the negative log likelihood (NLL) w.r.t. θ

Maximum Likelihood Estimation (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} p(\mathcal{D} \mid \theta)$$

$$= \arg\max_{\theta} \log p(\mathcal{D} \mid \theta)$$

$$= \arg\min_{\theta} \left[-\log p(\mathcal{D} \mid \theta) \right]$$

(Negative) Log Likelihood is Akin to Loss Function

For i.i.d. data $\mathcal{D} = \{y_1, \ldots, y_N\}$, using the log likelihood

$$\log p(\mathcal{D} \mid \theta) = \log \prod_{n=1}^{N} p(y_n \mid \theta)$$

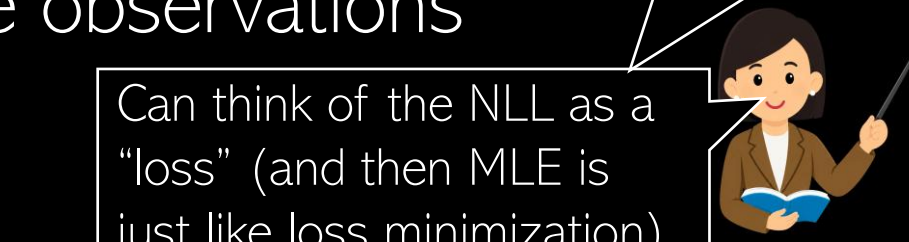
This also shows that the probabilistic viewpoint of learning from data is related to the non-probabilistic viewpoint

.. converts the product into a sum over all the observations

$$\log p(\mathcal{D} \mid \theta) = \sum_{n=1}^{N} \log p(y_n \mid \theta)$$

Using log often also helps prevent numerical underflow (product of many probabilities)

Can think of the NLL as a "loss" (and then MLE is just like loss minimization)



The general form of the MLE problem then becomes

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \sum_{n=1}^{N} \log p(y_n \mid \theta) = \arg\min_{\theta} \left[-\sum_{n=1}^{N} \log p(y_n \mid \theta) \right] = \arg\min_{\theta} \text{NLL}(\mathcal{D}; \theta)$$

maximum likelihood $\Leftrightarrow$ negative log–likelihood (NLL) minimization

MLE for Bernoulli Model

For the Bernoulli model, the MLE will be

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \sum_{n=1}^{N} \log p\,(y_n \mid \theta)$$

$$= \arg\max_{\theta} \sum_{n=1}^{N} \log[\theta^{y_n}(1-\theta)^{1-y_n}]$$

$$= \arg\max_{\theta}\ [N_1 \log\theta + N_0 \log(1-\theta)]$$

$N_1 = \sum_{n=1}^{N} y_n$, Number of successes $(y_n = 1)$

$N_0 = N - N_1$ Number of failures $(y_n = 0)$

Taking derivatives and setting to zero gives

$$\hat{\theta}_{\text{MLE}} = \frac{N_1}{N_0 + N_1} = \frac{N_1}{N}$$

Thus MLE in the Bernoulli case is simply the observed fraction of positives (successes) recorded in the dataset $\mathcal{D}$

Our manufacturing plant example: fraction of non-defective items

Issue with MLE: Overconfidence (Overfitting)

Consider a small sample with all defective items

$$\mathcal{D} = \{y_1, y_2, y_3, y_4, y_5\} = \{0, 0, 0, 0, 0\}$$

$$N_1 = 0 \Rightarrow \hat{\theta}_{\text{MLE}} = \frac{N_1}{N} = \frac{0}{5} = 0$$

This is problematic. If we pick a new "test" item randomly for inspection

$$p(y_* = 1 \mid \hat{\theta}_{\text{MLE}}) = 0$$

A direct consequence of MLE's overfitting on the training data $\mathcal{D} = \{0, 0, 0, 0, 0\}$

Important: Sample (dataset) size can also change reliability of MLE

$\text{MLE on } \mathcal{D}_A \text{ (0 positives out of 5)}$

$\text{MLE on } \mathcal{D}_B \text{ (0 positives out of 5000)}$

Even though $\hat{\theta}_{\text{MLE}} = 0$, this estimate is more reliable than $\hat{\theta}_{\text{MLE}} = 0$ obtained using $\mathcal{D}_A$

Reducing MLE's Overconfidence

In the non-probabilistic approaches to machine learning, we usually prevent overconfidence (overfitting) using a regularizer on parameters

We can try the same in the probabilistic approach and do

This is like a "regularized" MLE

$$\hat{\theta}_{\text{reg}} = \arg\min_{\theta} \left[-\sum_{n=1}^{N} \log p\left(y_n \mid \theta\right) + \lambda R(\theta) \right]$$

Here $R(\theta)$ is an appropriately chosen function that encourages θ to take "reasonable" values

Notion of what is a "reasonable" $R(\theta)$ depends on the model

For Bernoulli model, if we believe true θ to be close to some θ_* then $R(\theta) = |\theta - \theta_*|$ or $R(\theta) = (\theta - \theta_*)^2$ are some reasonable choices

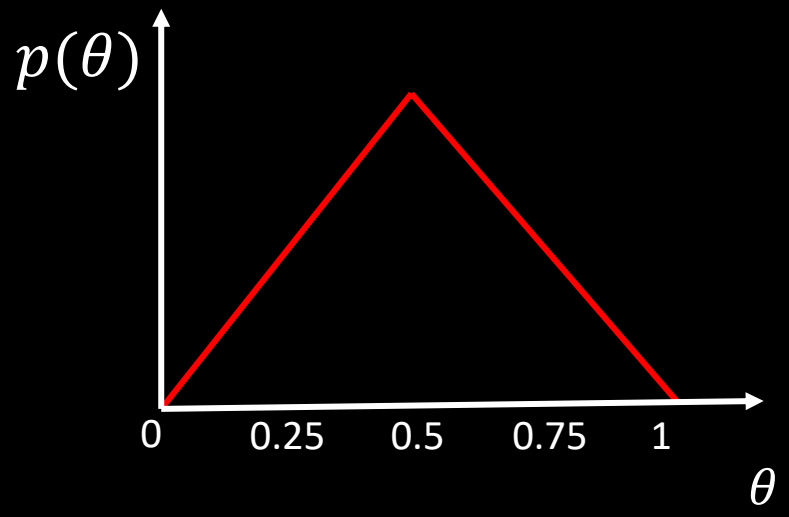
A more principled way to do this is to use a "prior distribution" over θ

Prior Distribution

A prior distribution $p(\theta)$ can be used to encode our beliefs about the possible values of θ can take before we have seen any actual data

For $\theta \in [0,1]$ here is a simplistic prior

Simplistic but not the most natural or mathematically convenient one – we will see a more natural prior for $\theta \in [0,1]$ shortly



What this prior expresses

It says that, *a priori*, we believe $\theta = 0.5$ is the most probable value

It says that, *a priori*, we believe the probability of θ being away from 0.5 linearly decreases on either side of 0.5

Prior Distribution to Posterior Distribution

Prior $p(\theta)$ expresses our prior belief: what the unknown θ might "look like" before we have seen seeing the data $\mathcal{D}$

Posterior $p(\theta|\mathcal{D})$ is updated belief about θ after we have seen $\mathcal{D}$

The Bayes' rule readily gives us the formula to compute the posterior

If we have two quantities A and B then Bayes' rule says

$$p(B \mid A) = \frac{p(A \mid B)p(B)}{p(A)}$$

Proof of Bayes' rule uses the simple fact from probability that

$$p(A, B) = p(A|B)p(B) = P(B|A)p(B)$$

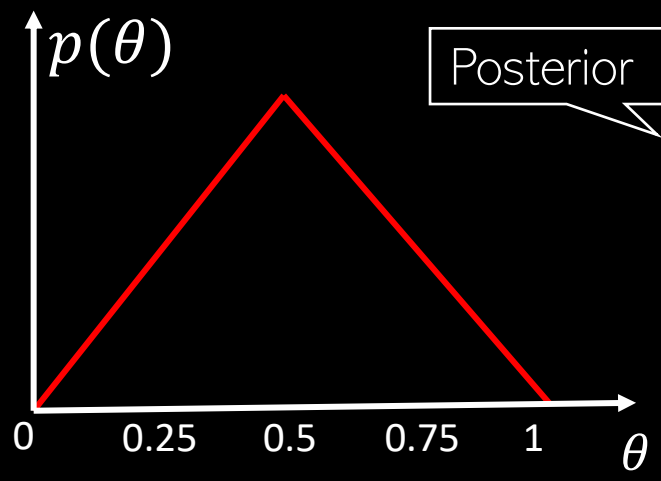


Posterior Distribution

Computing and using the full posterior distribution is known as Bayesian Machine Learning



The Bayes' rule readily gives us the formula to compute the posterior

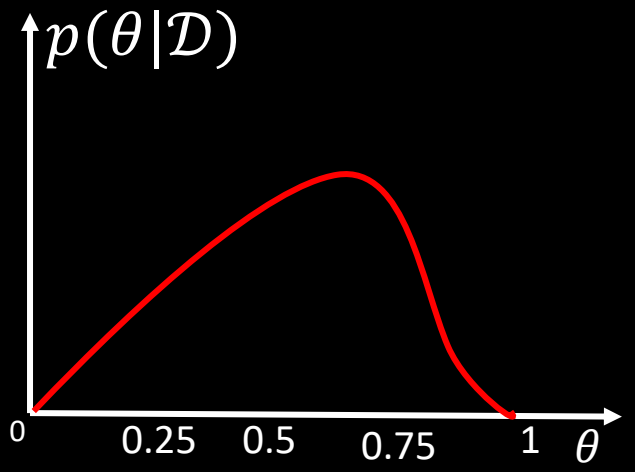


Likelihood

Prior

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

$$\boxed{p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta)}$$



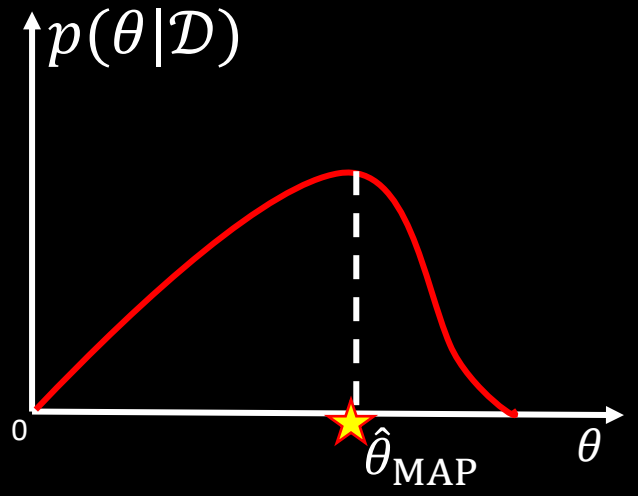
Note: The quantity $p(\mathcal{D}) = \int p(\mathcal{D}, \theta)d\theta = \int p(\mathcal{D} \mid \theta)p(\theta)d\theta$ is called the "marginal likelihood" (doesn't depend on θ)

In many cases (we will see some), posterior can be computed without the $p(\mathcal{D})$ (though $p(\mathcal{D})$ an important quantity in its own right)

Posterior Captures Useful Info for Estimated $\boldsymbol{\theta}$

We now have a distribution over possible values of $\boldsymbol{\theta}$ instead of a point estimate (like MLE)

Spread/variance of the posterior distribution tells us how confident we are about the estimated $\boldsymbol{\theta}$



Using posterior, we can also compute other useful quantities, e.g., probability that true $\boldsymbol{\theta}$ lies in a specific range

Even the single best value from the posterior is more robust than MLE

The maximum-a-posteriori (MAP) estimate (mode of the posterior)

$$\hat{\theta}_{\mathbf{MAP}} = \arg\max_{\theta} p(\theta \mid \mathcal{D})$$

Sort of a "regularized" version of MLE - justification on the next slide

MAP is akin to a Robust/Regularized MLE

Recall that

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \log p(\mathcal{D} \mid \theta) = \arg\min_{\theta} \text{NLL}(\mathcal{D}; \theta)$$

Now consider the MAP estimation problem

MAP still remains a point estimate but usually more robust than MLE



$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} p(\theta \mid \mathcal{D}) = \arg\max_{\theta} \log p(\theta \mid \mathcal{D})$$

$$= \arg\max_{\theta} \log \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}$$

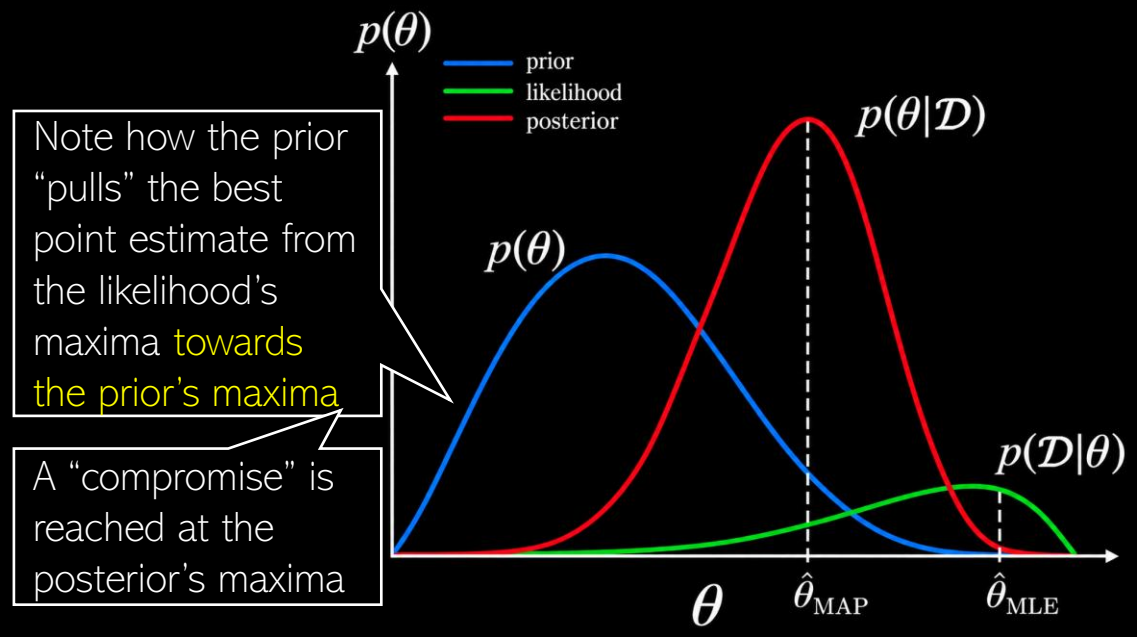
$$= \arg\max_{\theta} \log [p(\mathcal{D} \mid \theta)p(\theta)]$$

$$= \arg\max_{\theta} [\log p(\mathcal{D} \mid \theta) + \log p(\theta)]$$

$$= \arg\min_{\theta} [\text{NLL}(\mathcal{D}; \theta) - \log p(\theta)]$$

The regularization hyperparameter is part of the prior itself

The $\log p(\theta)$ term acts as a regularizer



Note how the prior "pulls" the best point estimate from the likelihood's maxima towards the prior's maxima

A "compromise" is reached at the posterior's maxima

Bernoulli Observation Model: MAP Estimate

Recall the likelihood - product of N Bernoullis – which had the form

$$p(\mathcal{D} \mid \theta) = \theta^{N_1}(1-\theta)^{N_0}$$

To compute MAP/posterior, we need a prior $p(\theta)$ over $\theta \in [0,1]$

Let's pick $p(\theta) = \mathbf{Beta}(\theta|\alpha,\beta)$

$$p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}$$

MAP objective will be: $\log p(\mathcal{D} \mid \theta) + \log p(\theta)$

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} [N_1 \log\theta + N_0 \log(1-\theta) + (\alpha-1)\log\theta + (\beta-1)\log(1-\theta)]$$

$$\Rightarrow \boxed{\hat{\theta}_{\text{MAP}} = \frac{N_1 + \alpha - 1}{N + \alpha + \beta - 2}}$$

Note how prior's hyperparams α, β act as additional "pseudo-observations" to make the MLE solution $\hat{\theta}_{\text{MLE}} = \frac{N_1}{N}$ "smooth" and reduce overfitting for small N_1 or N

